

AlphaProof: Anytime-Valid Certification for Agentic Alpha Discovery

Anonymous submission

Abstract

Autonomous quantitative-research agents repeatedly generate factors, inspect backtests, revise hypotheses, and decide when a result is convincing. This workflow has two statistical failure modes: search inflates the best score on a reused research sample, and repeated monitoring invalidates fixed-horizon tests even after a strategy is frozen. We introduce *AlphaProof*, a prospective certification layer that separates a search clock from an evidence clock. A candidate is registered before certification outcomes are revealed; bounded net utility then updates a predictable betting e-process. We prove candidate-level validity at arbitrary stopping times and family-wise error control for an unknown, adaptively arriving candidate stream without independence assumptions. On exact null markets, feedback-driven search raises the best research statistic from 1.04 to 4.11 as its budget grows from 4 to 256, while sealed performance remains near zero. Repeated t-tests falsely promote 33.3% of null streams and a naive 20-candidate agent promotes 94.0%; AlphaProof reduces these rates to 2.8% and 1.6% at a nominal 5% level. Calibration remains between 2.6% and 3.1% across Gaussian, heavy-tailed, GARCH, and regime-volatility nulls. In rolling U.S.-equity replays with costs, naive monitoring declares 40 of 50 candidates at least once, whereas AlphaProof certifies one. AlphaProof does not repair leaked backtests; it turns post-registration market outcomes into auditable, anytime-valid evidence.

Introduction

Large language models are turning quantitative research from a manually curated sequence of experiments into a closed feedback loop. Recent systems generate financial hypotheses, translate them into executable factors, run market backtests, store successful and failed trials, and use the resulting scores to choose the next experiment (Tang et al. 2025; Li et al. 2025; Wang et al. 2026; Shi et al. 2026). Stronger reasoning, memory, and search can therefore increase the number and adaptivity of specifications examined against the same historical market.

This progress changes the unit of statistical analysis. A conventional backtest treats a strategy as fixed before evaluation. An agentic researcher instead selects a strategy *because*

of earlier evaluations, may inspect a nominal holdout repeatedly, and may stop as soon as a favorable result appears. The final reported strategy hides both the rejected trials and the stopping rule. Financial econometrics has long recognized data snooping and factor proliferation (White 2000; Hansen 2005; Bailey and López de Prado 2014; Harvey, Liu, and Zhu 2016; Chordia, Goyal, and Saretto 2020); adaptive agents make the candidate universe path-dependent and potentially unbounded.

Two distinct failures are often conflated. First, repeated search on a research sample makes its best visible score optimistic. A new correction cannot turn that contaminated score into prospective evidence. Second, after a policy is frozen, a fixed-horizon p-value is still invalid if the agent monitors it after every outcome and stops opportunistically. The first problem demands data isolation or search-level diagnostics. The second demands inference that is valid uniformly over time. AlphaProof addresses the second while making the boundary explicit: an outcome observed before candidate registration can guide search, but can never certify that candidate.

We model autonomous discovery with two clocks. The *search clock* indexes adaptively generated candidates. The *evidence clock* indexes new market outcomes revealed only after registration. For a predeclared bounded net utility, AlphaProof maintains a nonnegative betting supermartingale, or e-process (Ramdas et al. 2023). Evidence may be inspected continuously, and the candidate may be stopped at any data-dependent time. For multiple candidates, a machine-readable evidence budget controls the probability of any false certification even when candidate birth times and policies depend on all previous information.

Our contributions are:

- We formalize *agentic alpha certification* as a two-clock, filtration-aware problem and distinguish reused research feedback from admissible post-registration evidence.
- We introduce a bounded predictable-betting certificate and prove anytime candidate validity and global family-wise error control for adaptively arriving candidates under arbitrary cross-candidate dependence.
- We build an executable benchmark spanning search inflation, optional stopping, candidate multiplicity, heavy tails, volatility dynamics, planted signals, and five rolling

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real-market windows. The experiments show both false-discovery control and the power–delay price of credible evidence.

Related Work

Agentic quantitative research. AlphaAgent regularizes originality, hypothesis alignment, and expression complexity to counter alpha decay (Tang et al. 2025). RD-Agent(Q) connects factor and model development through evaluation feedback and a bandit scheduler (Li et al. 2025). FactorMiner accumulates reusable skills and experience memory, while Hubble feeds family-level diagnostics back into formula generation (Wang et al. 2026; Shi et al. 2026). These systems aim to improve search quality, safety, or efficiency. AlphaProof is complementary: it treats the generator as a black box and asks what post-search evidence can support a statistical claim.

Backtest overfitting and adaptive analysis. White’s Reality Check and Hansen’s SPA test evaluate the best model in a tested universe (White 2000; Hansen 2005). The Deflated Sharpe Ratio (DSR) adjusts for selection, sample length, and non-normal returns, and combinatorial cross-validation estimates the probability of backtest overfitting (Bailey and López de Prado 2014; Bailey et al. 2017). These methods remain important diagnostics, but they require a recoverable trial universe or resampling design. In our null experiment, count-based Bonferroni is calibrated for random candidate search but fails when evolutionary proposals condition on the same scores. The broader adaptive-data-analysis literature similarly shows that repeated holdout access breaks generalization (Dwork et al. 2015; Russo and Zou 2016). Recent financial work proposes falsification on induced null environments (Nikolopoulos 2026). AlphaProof takes the stricter prospective route: no statistic computed before registration is reused as certification evidence.

Anytime-valid inference. E-values and e-processes remain valid under continuous monitoring and optional stopping (Ramdas et al. 2023; Howard et al. 2021). E-BH controls false discovery under arbitrary dependence, and e-LOND extends e-values to online hypothesis streams (Wang and Ramdas 2022; Xu and Ramdas 2024); related p-value methods control online FDR under stronger dependence conditions (Javanmard and Montanari 2018). In finance, e-backtesting has been developed for VaR and Expected Shortfall forecasts (Wang, Wang, and Ziegel 2026). Our setting differs in that hypotheses are executable trading policies born from an adaptive AI search process, costs and risk preferences enter the tested utility, and both market time and candidate time must be recorded.

Problem Formulation

Let $(\mathcal{F}_t)_{t \geq 0}$ contain all market observations revealed through time t , all prior research feedback, candidate registrations, and the agent’s internal state. Candidate j is born at an $\mathcal{F}_t$ -stopping time b_j . Its position $w_{j,t-1}$ may adapt to information after birth, but must be $\mathcal{F}_{t-1}$ -measurable: the position is committed before return r_t is revealed.

Raw P&L is not automatically the scientific estimand. We predeclare

$$\begin{aligned} R_{j,t}^{\text{net}} &= w_{j,t-1}^\top r_t - c(\Delta w_{j,t-1}) - r_t^{(b)}, \\ X_{j,t} &= g(R_{j,t}^{\text{net}}, z_t), \quad |X_{j,t}| \leq B, \end{aligned} \quad (1)$$

where c includes trading frictions, $r_t^{(b)}$ is a benchmark, z_t may contain risk state, and g is a bounded utility fixed at registration. If clipping is used, AlphaProof certifies clipped utility rather than unrestricted profitability.

Definition 1 (Candidate null) *Candidate j has no positive conditional net utility when*

$$H_j : \quad \mathbb{E}[X_{j,t} \mid \mathcal{F}_{t-1}] \leq 0 \quad \text{for every } t > b_j. \quad (2)$$

The goal is to certify candidates while allowing: (i) arbitrary internal search before b_j ; (ii) continuous monitoring; (iii) data-dependent stopping; (iv) an unknown number of candidates; and (v) dependent candidate returns driven by the same market. Look-ahead leakage violates the filtration and is outside the claim.

AlphaProof

Registration and Predictable Betting

For normalized utility $U_{j,t} = X_{j,t}/B \in [-1, 1]$, choose a betting fraction $\lambda_{j,t} \in [0, 1]$ using only $\mathcal{F}_{t-1}$ and update

$$E_{j,t} = E_{j,t-1}(1 + \lambda_{j,t}U_{j,t}), \quad E_{j,b_j} = 1. \quad (3)$$

Our implementation uses the plug-in rule

$$\lambda_{j,t} = \text{clip}\left(\frac{S_{j,t-1}}{Q_{j,t-1} + \eta}, 0, \lambda_{\max}\right), \quad (4)$$

where S and Q are the past sum and squared sum of U , $\eta = 1$, and $\lambda_{\max} = 0.95$. This approximates a one-sided Kelly fraction while remaining predictable. We also implement a fixed mixture of constant betting fractions; both constructions are valid, but the plug-in update is faster and avoids a fixed mixture penalty.

Theorem 1 (Anytime candidate validity) *Under H_j , predictability, and $|X_{j,t}| \leq B$, $(E_{j,t})_{t \geq b_j}$ is a nonnegative supermartingale with initial value one. For every possibly data-dependent stopping time τ_j ,*

$$\mathbb{E}[E_{j,\tau_j}] \leq 1, \quad \mathbb{P}\left(\sup_{t > b_j} E_{j,t} \geq \alpha_j^{-1}\right) \leq \alpha_j. \quad (5)$$

Proof. The factor in Eq. (3) is nonnegative. Because $\lambda_{j,t}$ is $\mathcal{F}_{t-1}$ -measurable, $\mathbb{E}[1 + \lambda_{j,t}U_{j,t} \mid \mathcal{F}_{t-1}] \leq 1$ under H_j . Iterated conditioning gives the supermartingale property. Optional stopping and Ville’s inequality give Eq. (5). $\square$

The theorem is architecture-agnostic: the candidate may come from an LLM, evolutionary program, reinforcement learner, or human, because validity depends on information order rather than generator complexity.

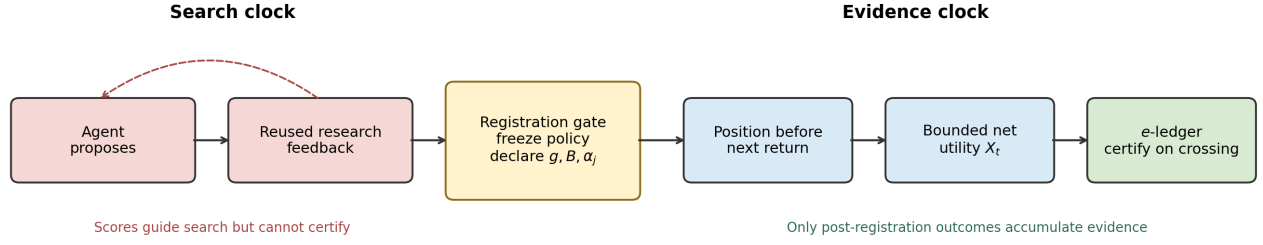


Figure 1: AlphaProof separates unrestricted research feedback from admissible prospective evidence. The registration gate freezes the policy interface, bounded utility, and testing allocation. Only subsequently revealed outcomes update the e-ledger.

Algorithm 1 AlphaProof Evidence Ledger

Require: Global budget α , utility bound B

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1: for each market time  $t$  do
2:   for each new candidate  $j$  proposed from  $\mathcal{F}_{t-1}$  do
3:     Freeze interface and register  $(g, B, b_j, \alpha_j)$ 
4:     Set  $E_{j,b_j} = 1, S_{j,b_j} = Q_{j,b_j} = 0$ 
5:   end for
6:   for each live candidate  $j$  do
7:     Commit  $w_{j,t-1}$ ; then reveal  $r_t$ 
8:     Compute  $X_{j,t}$  by Eq. (1)
9:     Choose predictable  $\lambda_{j,t}$  by Eq. (4)
10:    Update  $E_{j,t}$  by Eq. (3)
11:    if  $E_{j,t} \geq 1/\alpha_j$  then
12:      Certify  $j$  and record the first crossing
13:    end if
14:  end for
15: end for

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The Evidence Ledger

Before candidate j observes its evidence, assign a predictable budget $\alpha_j > 0$ such that $\sum_j \alpha_j \leq \alpha$. A finite registered family may use equal or research-prior-weighted allocation. For an unknown infinite stream, we use

$$\alpha_j = \frac{6\alpha}{\pi^2 j^2}. \quad (6)$$

Candidate j is certified at its first crossing of $1/\alpha_j$.

Theorem 2 (Adaptive-family FWER control) *Let $\mathcal{H}_0$ index true candidate nulls. Candidate identities, birth times, policies, and stopping times may depend on the past, and candidate e-processes may be arbitrarily dependent. Then*

$$\mathbb{P}(\exists j \in \mathcal{H}_0 : j \text{ is certified}) \leq \sum_{j \in \mathcal{H}_0} \alpha_j \leq \alpha. \quad (7)$$

Proof. Condition on $\mathcal{F}_{b_j}$ and apply Theorem 1 to each true null. The union bound yields Eq. (7); independence is not required. $\square$

Under a stationary alternative with $\mathbb{E}[\log(1 + \lambda U_{j,t})] = g_\lambda > 0$, the strong law gives $t^{-1} \log E_{j,t} \rightarrow g_\lambda$; crossing therefore occurs almost surely and its first-order delay is $\log(1/\alpha_j)/g_\lambda$. This exposes the price of validity: weaker

utility or smaller candidate budgets require more evidence time.

Experiments

We evaluate four questions. **RQ1:** Does adaptive search manufacture increasingly strong results under an exact null? **RQ2:** Does AlphaProof control false promotion under optional stopping, dependent candidate streams, heavy tails, and volatility dynamics? **RQ3:** What power and delay does valid certification incur? **RQ4:** How does the protocol behave in rolling real-market replay?

Synthetic Protocol

Search inflation. We generate 16 persistent exogenous features and symmetric bounded martingale-difference returns. Random search evaluates independent coefficient vectors. Evolutionary search retains high-scoring parents and mutates their coefficients using feedback from the same 252-day adaptation sample. From budgets $K \in \{4, 16, 64, 256\}$, the best candidate is evaluated on a disjoint 252-day locked sample. We repeat 250 times and report studentized net payoff. On the reused adaptation sample we compare a naive one-sided test, count-based Bonferroni, and DSR.

Sequential calibration. We simulate 3,000 Gaussian null streams of 756 observations. A fixed-horizon t-test is evaluated once; an invalid repeated t-test is inspected after every observation from day 20; Time-Bonferroni divides α by the number of looks; AlphaProof is monitored continuously. Robustness uses 1,000 streams each from Gaussian, Student- t_4 , GARCH, and two-state regime-volatility nulls. All are symmetrically bounded and satisfy Eq. (2).

Adaptive candidate stream and power. An agent fits a ridge-linear feature policy to all revealed history and spawns a new frozen candidate every 25 observations, producing 20 candidates over 600 steps. We compare “any candidate, any time” $p \leq .05$ with the ledger in Eq. (6) across 500 null runs. For power, returns follow $r_t = \tanh(\delta z_{t,1} + \epsilon_t)$; we use 180 runs per signal strength, a 900-step horizon, and report certification probability and median crossing time. All tests use $\alpha = .05$. Confidence intervals are Wilson intervals.

Rolling U.S.-Equity Replay

We use a public daily OHLCV panel of 200 U.S. stocks, 799,101 rows, spanning 1962–2026 (younginpiniti 2026). Five nonoverlapping evidence windows follow eight-year research windows. For each split, the 100 most liquid stocks with at least 80% research-window coverage and median price above 5 form the universe; the available count ranges from 55 to 89. Seven strictly lagged, cross-sectionally ranked features capture one-day reversal, 5/20/60-day momentum, 20-day volatility, range, and volume surprise.

Ten research agents independently search 128 random linear combinations on the research window and freeze their best policy before evidence begins. Each policy forms a dollar-neutral portfolio. Net payoff subtracts 10 basis points per unit turnover and the registered utility clips daily net return to $[-5\%, 5\%]$. The known ten-candidate family receives equal $\alpha/10$ budgets. The fixed source universe may contain survivorship bias; accordingly, this study tests operational behavior, not market efficiency or deployable profitability.

Results

Search Produces Scores, Not Evidence

Figure 2 and Table 1 show monotone null inflation. For evolutionary search, the best adaptation statistic grows from 1.04 at $K = 4$ to 4.11 at $K = 256$, while the locked mean remains 0.07. Random search follows the expected extreme-value pattern; feedback-driven evolution grows faster because later candidates explicitly fit earlier scores.

At $K = 256$, naive adaptation-sample discovery reaches 98.0% for random search and 100% for evolutionary search. Bonferroni controls random search at 3.2% but yields 83.2% false discoveries for evolution: counting proposals does not repair candidate p-values whose construction already depends on the sample. DSR is conservative here (0% random, 0.4% evolutionary). AlphaProof does not claim to salvage these adaptation statistics; it begins at registration.

Anytime and Family-Level Calibration

Figure 3 isolates optional stopping from ordinary fixed-horizon inference. The fixed t-test is calibrated at 5.3%, but continuous peeking raises false promotion to 33.3%. Time-Bonferroni is valid but highly conservative (0.2%); AlphaProof is 2.8% (95% CI 2.3–3.5%). In the adaptive 20-candidate stream, naive promotion reaches 94.0% (91.6–95.8%), whereas the ledger yields 1.6% (0.8–3.1%).

The guarantee is not a Gaussian artifact. Table 2 shows AlphaProof false-promotion rates of 2.6–3.1% across all four null families. Repeated t-tests remain between 32.5% and 34.6%. Conservative rates are expected because Ville’s inequality is uniform over all times and the plug-in bet starts cautiously.

Power–Delay Tradeoff

Table 3 confirms that safety is not vacuous. At $\delta = .20$, AlphaProof certifies 63.3% of runs (56.1–70.0%) with median delay 545. At $\delta = .35$, all runs certify and median delay falls to 238; stronger signals cross earlier. The naive rule is

nearly always positive even at $\delta = 0$, so its apparent power is inseparable from its false discoveries.

Real-Market Replay

Table 4 shows why prospective certification changes the conclusion drawn from a backtest. Across five windows, fixed-horizon tests call 25 of 50 candidates significant; continuous naive monitoring calls 40. AlphaProof certifies one candidate, in 2019–2020, despite positive mean locked statistics in every window. The result should not be read as “only one strategy is profitable.” It means only one candidate accumulates enough bounded net evidence to clear its family-adjusted, anytime-valid threshold under the stated utility. Evidence requirements are intentionally stronger than ranking by annualized Sharpe.

Discussion and Limitations

AlphaProof changes the interface between an autonomous researcher and a market evaluation service. Search feedback remains rich and unrestricted, but every claim exposes its birth time, policy version, utility, bound, cost model, evidence allocation, and first crossing. This supports audit after a model, prompt, or agent architecture changes. It also makes negative results useful: failure to cross is not proof of no predictability, but an explicit statement that available evidence is insufficient for the registered claim.

The guarantee has sharp boundaries. First, AlphaProof cannot detect a policy that secretly reads future data; timing and data lineage must be enforced by the execution system. Second, bounded utility may differ from investor utility, and ex-post changes to g or B create a new candidate. Third, the conditional null is strong but interpretable: certification concerns the realized policy relative to its filtration, benchmark, and costs, not an invariant economic law. Fourth, FWER is conservative for large streams. E-BH and e-LOND suggest more powerful online FDR extensions (Wang and Ramdas 2022; Xu and Ramdas 2024), but combining candidate births, persistent shared market dependence, and policy replacement deserves separate analysis. Finally, the real panel has a fixed 200-stock source universe, uncertain delisting coverage, and simplified transaction costs. Synthetic nulls, where truth is known, remain the primary calibration evidence.

Broader impact. AlphaProof is an evaluation method, not an automated trading recommendation. It may reduce false confidence from large-scale AI search, but a certificate does not address capacity, market impact, legal suitability, tail events beyond the utility bound, or future regime change. Deployment still requires human risk ownership and independent operational controls.

Conclusion

Autonomous agents can increase the depth of quantitative search without increasing the information contained in a historical market. The resulting backtest score is a search output, not evidence by itself. AlphaProof creates a clean evidence clock: register first, act before outcomes, accumulate bounded net utility through an e-process, and spend a

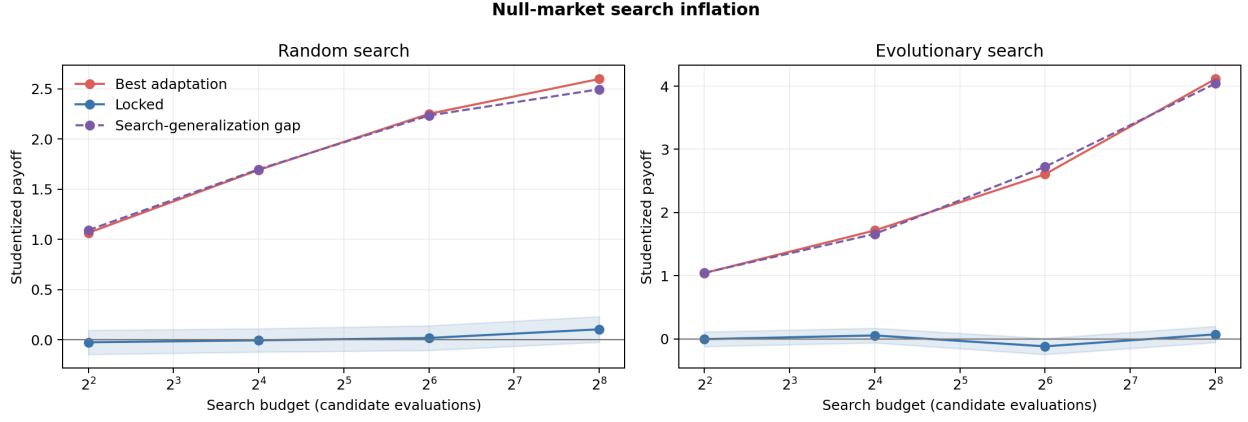


Figure 2: In exact null markets, the best adaptation score and the search–generalization gap rise with search budget; the selected candidate’s locked score remains centered near zero. Shading is a 95% interval for the mean locked score over 250 repetitions.

Search	K	Adapt.	Locked	Gap	Naive	Bonf.	DSR
Evolutionary	4	1.04	-0.00	1.05	17.2	6.8	0.0
Evolutionary	16	1.72	0.05	1.66	50.0	4.4	0.0
Evolutionary	64	2.61	-0.12	2.72	96.8	14.4	0.0
Evolutionary	256	4.11	0.07	4.04	100.0	83.2	0.4
Random	4	1.06	-0.03	1.09	20.8	6.8	0.0
Random	16	1.69	-0.01	1.70	50.8	1.2	0.0
Random	64	2.25	0.02	2.23	91.6	2.8	0.0
Random	256	2.60	0.10	2.50	98.0	3.2	0.0

Table 1: Null search results. Discovery columns are percentages computed on the reused adaptation sample. AlphaProof treats these columns as diagnostics, not certification evidence.

Method	Gaussian	t_4	GARCH	Regime-vol.
Fixed t	5.5	4.5	6.1	3.8
Repeated t	33.2	34.6	32.5	33.5
Time-Bonf.	0.1	0.3	0.2	0.6
AlphaProof	2.9	2.7	2.6	3.1

Table 2: False-promotion percentages across null families (1,000 runs each).

Signal δ	Naive	AlphaProof	Median delay
0.00	97.2	1.1	384
0.20	100.0	63.3	545
0.35	100.0	100.0	238
0.50	100.0	100.0	167
0.70	100.0	100.0	138

Table 3: Promotion percentages and evidence-clock delay.

visible error budget across candidates. The resulting guarantees survive optional stopping, adaptive candidate arrival, and shared market dependence. Controlled nulls and rolling-market replay show a consistent empirical lesson: credible agentic finance requires not only better factor generation, but a validation interface that the agent cannot silently train on.

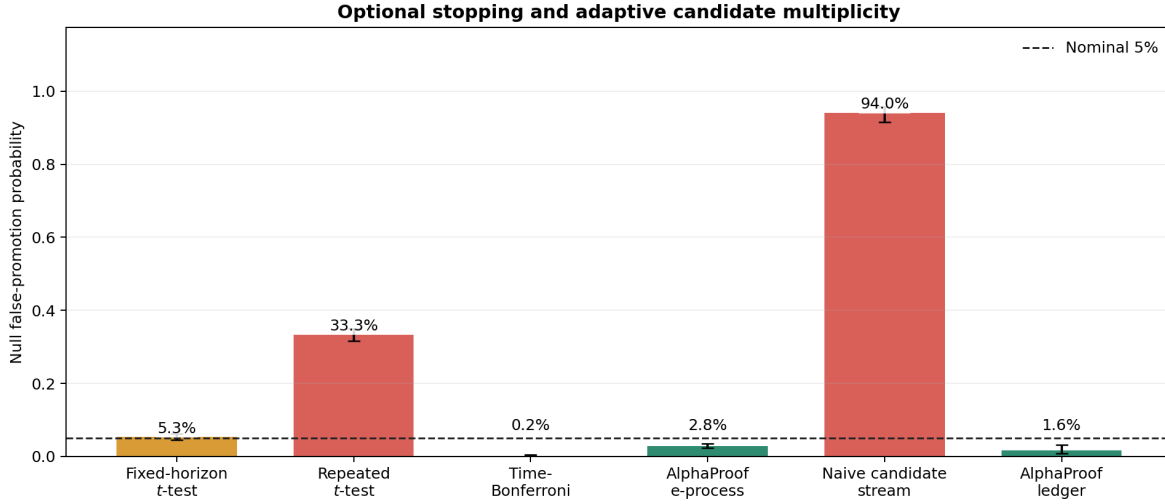


Figure 3: Null false-promotion probability. Error bars are 95% Wilson intervals; the dashed line is the nominal 5% level.

Evidence	N	Adapt.	Locked	Fixed	Peek	Proof
2013–2014	55	7.94	1.44	0	10	0
2016–2017	62	1.80	1.94	7	8	0
2019–2020	73	1.47	2.39	8	9	1
2022–2023	81	1.52	1.64	7	7	0
2024–2026	89	6.96	1.36	3	6	0

Table 4: Rolling-origin replay. Each window has ten registered candidates. Fixed and Peek count naive fixed-horizon and anytime significance; Proof counts AlphaProof certifications.

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